

Rx controller Functional Design Specifications

UCle 3.0 Physical Layer

Version 1

Block Owner

Authors

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1 Table of Contents

1	Table of Contents	1
2	Revision History	2
3	Overview	3
4	Operation and Description.....	3
4.1	Digital Interface	3
4.1.1	Parameters Names	3
4.1.2	Ports Names	3
4.2	Functional Description.....	4
	RX FSM Main State Entry Encodings	8
4.3	Block Diagram.....	9

2 Revision History

Version	Date	Author(s)	Revision Notes	Owner Approval

3 Overview

This module is responsible for configuring each block in the RX datapath based on the LTSM state conditions.

4 Operation and Description

4.1 Digital Interface

4.1.1 Parameters Names

Parameter Name	Default	Description

4.1.2 Ports Names

Port Name	Port Width	Port Type	Description
i_clk	1	input	The module clk signal.
i_reset	1	input	The module reset signal. Active high Asynchronous reset.
i_lane_map_code	3	input	Lane map code from the ltism to identify
i_rx_encoding	9	input	The encoding comes from the ltism to indelicate the current state.
i_error_threshold	16	input	Error count threshold for compare-mode valid detection from LTSM.
i_fifo_empty	1	input	Flag from FIFO that indicates FIFO is empty
i_half_rate	1	input	Rate mode selector. [1] routes half-rate detector output [0] routes quadrature-rate detector output From LTSM to controller.
o_fifo_rd_en	1	output	Flag from FIFO to enables read
o_error_threshold	16	output	Error count threshold for compare-mode valid detection.
o_rx_lfsr_enable	1	output	Enables the rx lfsr block
o_rx_lfsr_load	1	output	Loads the seed values of the rx lfsr
o_rx_lfsr_train	1	output	Signal to identify if the data is destined to be descrambled or be compared by rx lfsr
o_pattern_type	2	output	Detection enable flags. Bit [1] enables valid detection Bit [0] enables clock detection.
o_detection_type	1	output	Valid detection algorithm selector. 0: selects per-lane (consecutive-correct) mode 1: selects compare/error-count mode
o_data_det_type	1	output	Select between lfsr pattern and per-lane id pattern. 1 : lfsr pattern. 0: per-lane id pattern.

o_halfrate	1	output	Rate mode selector. [1]routes half-rate detector output [0]routes quadrature-rate detector output From controller to clock valid detection block.
o_rx_done	1	output	Indicates that the datapath finishes its operation.
o_per_lane_id_det_enable	1	output	Enables the per-lane id generator block.
o_mb_clk_p_en	1	output	Enables the AFE mainband positive clock tri-state.
o_mb_clk_n_en	1	output	Enables the AFE mainband negative clock tri-state.
o_mb_valid_en	1	output	Enables the AFE mainband valid tri-state.
o_mb_track_en	1	output	Enable the AFE mainband track tri-state.
o_mb_lanes_en	16	output	Enables the AFE mainband lanes tri-state.
o_l2b_enable	1	output	Enables the b2l block while the data transmission.

4.2 Functional Description

1. LFSR configurations:

- LFSR must be enabled in pattern detection state in the RX initiated eye sweep test (0x18A) in all states required to do RX initiated eye sweep test using lfsr-pattern for data lanes.
- LFSR must be enabled in pattern detection state in the TX initiated eye sweep test (0x182) in all states required to do **RX** initiated eye sweep test using lfsr-pattern for data lanes.
- seed values must be loaded in both TX initiated eye sweep test (0x181) and RX initiated eye sweep test (0x189) regarding the state will use per-lane id pattern or lfsr pattern.

The following are the detailed states using the lfsr pattern for data lanes and the condition of the o_rx_lfsr_load and o_rx_lfsr_train signals.

- Seed values must be loaded to the LFSR in MBTRAIN.DATAVREF/RX initiated eye sweep test (0x189), and the train mode must be enabled here.
- Seed values must be loaded to the LFSR in MBTRAIN.DATATRAINCENTER1/RX initiated eye sweep test (0x189), and the train mode must be enabled here.
- Seed values must be loaded to the LFSR in MBTRAIN.DATATRAINVREF/RX initiated eye sweep test (0x189), and the train mode must be enabled here.

- Seed values must be loaded to the LFSR in MBTRAIN.RXDESKEW/RX initiated eye sweep test (0x189), and the train mode must be enabled here.
 - Seed values must be loaded to the LFSR in MBTRAIN.DATATRAINCENTER2/TX initiated eye sweep test (0x181), and the train mode must be enabled here.
 - Seed values must be loaded to the LFSR in MBTRAIN.LINKSPEED/TX initiated eye sweep test (0x181), and the train mode must be enabled here.
- d. After (1cycle) *128 the, o_rx_done is asserted in each state uses the lfsr pattern for data lanes.
- e. Seed values must be loaded to the LFSR in LINKINIT (0x100).
- f. Train mode will be disabled in ACTIVE state and enabled for each mentioned states using the lfsr pattern for data lanes.
- g. For states use the lfsr pattern the o_data_pattern_type must be asserted.
- h. For states use per-lane id pattern o_data_pattern_type must be zero (MBINIT.REVERSALMB / MBINIT REVERSAL TX Per Lane ID generation (0x34)) and (MBINIT.REPAIRMB / TX initiated eye sweep test (0x182)).

2. clk/valid/trackPattern Detector configurations.

o_pattern_type[1:0]	Valid detection	Clock detection	Description
00	Disabled	Disabled	No detection enabled
01	Disabled	Enabled	Clock detection enabled only
10	Enabled	Disabled	Valid detection enabled only
11	Enabled	Enabled	Both Valid and Clock detection enabled

- a. Clock-only settings must be applied to the detector in the MBINIT.REPAIRCLK / MBINIT REPAIRCLK RX Clk Pattern Detection (0x21). Once the pattern detection ends (after (16 + 8 cycles) * 128), the o_rx_done is asserted to indicate that the datapath finished the pattern detection.

- b. Valid-only pattern settings must be applied to the detector in the MBINIT.REPAIRVAL / MBINIT REPAIRVAL RX Valid Pattern Detection (0x29). Once the pattern detection ends (after $(4 + 4 \text{ cycles}) * 128$), the o_rx_done is asserted to indicate that the datapath finished the pattern detection.
- c. Valid-only pattern settings must be applied to the detector in the MBTRAIN.VALVREF / RX initiated eye sweep test (0x18A). Once the pattern detection ends (after $(4 + 4 \text{ cycles}) * 128$), the o_rx_done is asserted to indicate that the datapath finished the pattern detection.
- d. Active/data pattern settings must be applied to the detector in the MBTRAIN.VALTRAINCENTER / TX initiated eye sweep test (0x182). Once the pattern detection ends (after $(4 + 4 \text{ cycles}) * 128$), the o_rx_done is asserted to indicate that the datapath finished the pattern detection.
- e. Active/data pattern settings must be applied to the detector in the MBTRAIN.VALTRAINVREF / RX initiated eye sweep test (0x18A). Once the pattern detection ends (after $(4 + 4 \text{ cycles}) * 128$), the o_rx_done is asserted to indicate that the datapath finished the pattern detection.
- f. Active/data pattern settings must be applied to the detector in the MBINIT.REVERSALMB / MBINIT REVERSAL RX Per Lane ID generation (0x32). Once the pattern detection ends (after 128 cycles), the o_rx_done is asserted to indicate that the datapath finished the pattern detection.
- g. Active/data pattern settings must be applied to the detector in the MBINIT.REPAIRMB / TX initiated eye sweep test (0x182). Once the pattern detection ends (after 128 cycles), the o_rx_done is asserted to indicate that the datapath finished the pattern detection.
- h. Active/data pattern settings are applied for states using the lfsr data pattern mentioned above.
- i. Active/data pattern settings will be used in the Active state.

3. AFE Drivers configurations (note the lane held low or not is kept to the AFE driver modules but mentioned here to be reference to the driver modules designer):

- a. In RESET state the Data, Valid, Clock, Track and data lanes Receivers are disabled (their enables are zero).
- b. In SBINIT state the Data, Valid, Clock, Track and data lanes Receivers remain disabled.

- c. In MBINIT.PARAM and MBINIT.CAL states the Data, Valid, Clock, Track and data lanes Receivers remain disabled.
- d. In MBINIT.REPAIRCLK state the Clock and track lanes only are enabled while keeping the other lanes disabled.
- e. In MBINIT.REPAIRVAL state the Data, Valid, Clock, Track and data lanes Receivers enabled.
- f. In MBINIT.REVERSALMB state the Data, Valid, Clock, Track and data lanes Receivers enabled.
- g. In MBINIT.REPAIRMB state the Data, Valid, Clock, Track and data lanes Receivers enabled..
- h. In all MBTRAIN states the Data, Valid, Clock, Track and data lanes Receivers enabled.
- i. In LINKINIT state the Data, Valid, Clock, Track and data lanes Receivers enabled.
- j. In ACTIVE state the Data, Valid, Clock, Track and data lanes Receivers enabled.
- k. In PHYRETRAIN state the Data, Valid, Clock, Track and data lanes Receivers enabled.
- l. In TRAINERROR state the Data, Valid, Clock, Track and data lanes Receivers enabled.
- m. In L1/L2 the Data, Valid, Clock, Track and data lanes Receivers disabled.

note : after the width degradation (if possible), the functional lanes only that will be enabled at the states that we mentioned that the data lanes are enabled above .see the following table:

Table 4-9. Standard Package Logical Lane Map

Lane Map Code	Functional Lanes
000b	None (Degrade not possible)
001b	Logical Lanes 0 to 7
010b	Logical Lanes 8 to 15
011b	0 - 15
100b	Logical Lanes 0 to 3
101b	Logical Lanes 4 to 7

4. data transmission configurations (RDI interface)

- a. When lane to byte has valid data , o_pl_valid is asserted

5. Error handling

- a.

flow of operation

Module first samples the incoming i_rx_encoding each cycle and then compare it by local table of main Itsm states stored in it in enum called Itsm_states_e .

RX FSM Main State Entry Encodings

Main State	Binary Encoding [8:0]	Hex Encoding
RESET	00 0000 000	0x00
SBINIT	00 0001 000	0x08
MBINIT PARAM	00 0010 000	0x10
MBINIT CAL	00 0011 000	0x18
MBINIT REPAIRCLK	00 0100 000	0x20
MBINIT REPAIRVAL	00 0101 000	0x28
MBINIT REVERSAL	00 0110 000	0x30
MBINIT REPAIRMB	00 0111 000	0x38
MBTRAIN VALVREF	01 0000 000	0x80
MBTRAIN DATAVREF	01 0001 000	0x88
MBTRAIN DTC1	01 0010 000	0x90
MBTRAIN RXCLKCAL	01 0011 000	0x98
MBTRAIN VALTRAINVREF	01 0100 000	0xA0
MBTRAIN RXDESKEW	01 0101 000	0xA8
MBTRAIN DTC2	01 0110 000	0xB0
MBTRAIN LINKSPEED	01 0111 000	0xB8
MBTRAIN REPAIR	01 1000 000	0xC0
MBTRAIN SPEEDIDLE	01 1001 000	0xC8
MBTRAIN TXSELF CAL	01 1010 000	0xD0
PHYRETRAIN	01 1011 000	0xD8

TRAINERROR	01 1100 000	0xE0
VALTRAINCENTER	01 1101 000	0xE8
DATATRAINVREF	01 1110 000	0xF0
LINKINIT	10 0000 000	0x100
ACTIVE	10 0001 000	0x108
L1 / Exit HS	10 0010 000	0x110

The module then stores the current state in internal register called `ltsm_current_state`.

Based on this `current_state` value the controller can take decisions that depend on the two information of the current main state and the current `i_rx_encoding` (representing the current substate).

4.3 Block Diagram

